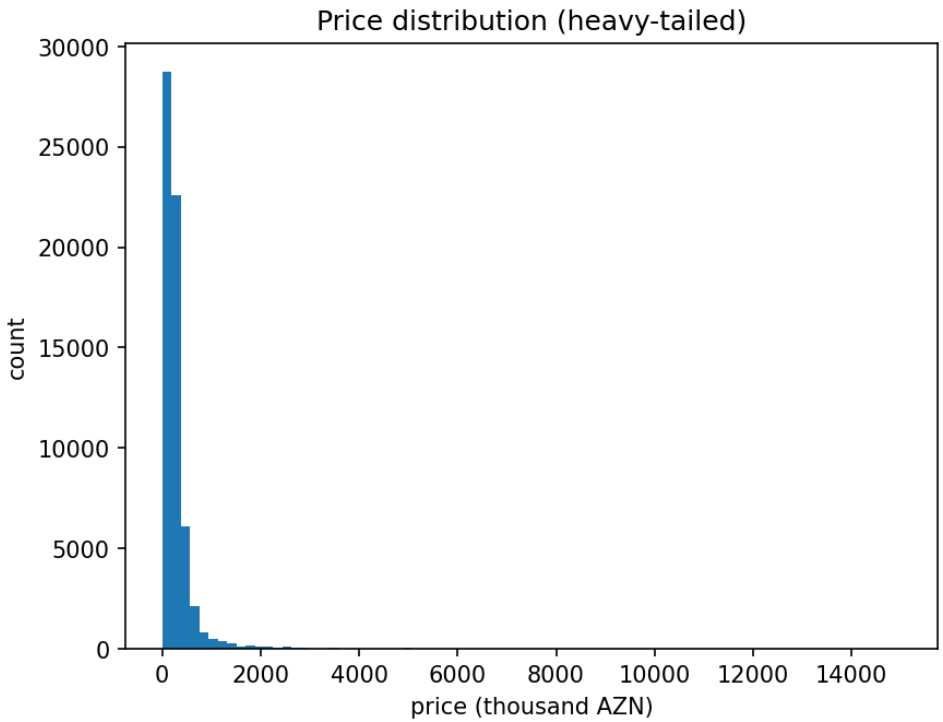
Bina.az Price Prediction: From-Scratch Trees and a Pegasos SVM

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Problem and data

- Task A: predict sale price (trained on $\log(\text{price})$).
- Task B: premium vs. standard tier at train median 213,000 AZN.
- 100,775 raw rows $\rightarrow$ 62,673 clean, 43 features, split 43869/9402/9402.

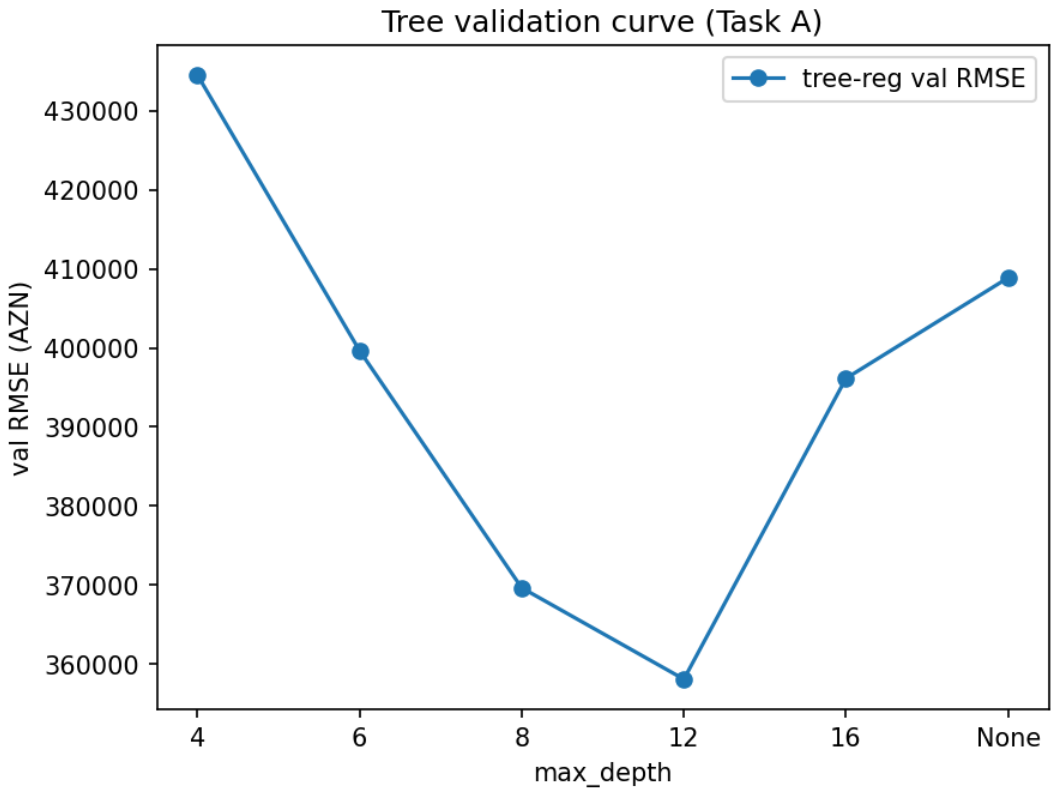


Cleaning and leakage

- Dedup on listing URL; parsed area, floors ("7 / 9"), land (sot).
- Dropped price/area outliers and off-map coordinates; all prices AZN.
- Removed `unit_price`, `total_price` (== price) + IDs, URLs, names, free text.
- Threshold and scaling statistics from train only.

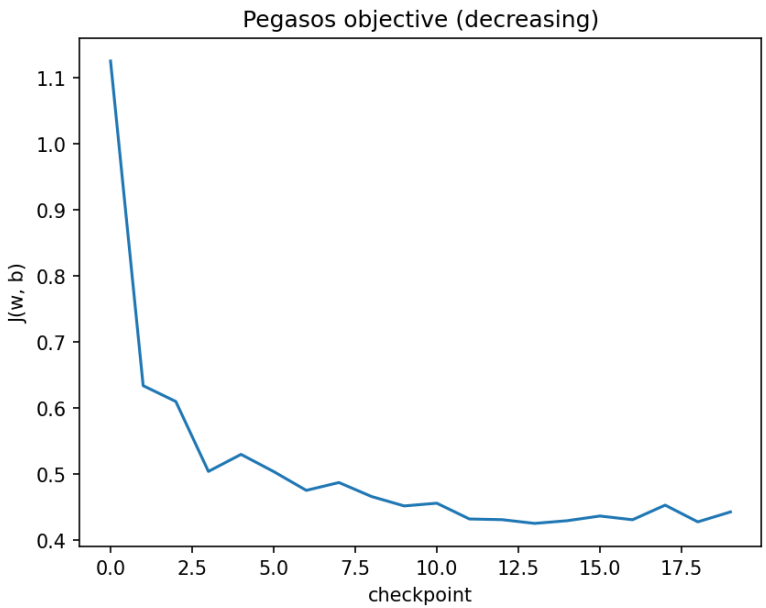
Decision tree (from scratch)

- Gini / entropy / MSE, exact split search, depth/leaf/impurity stops.
- Task A: best depth 12, leaf 5. Task B: Gini 0.918 beats entropy 0.916.



Pegasos SVM (from scratch)

- $J(w,b) = \frac{\lambda}{2}\|w\|^2 + \frac{1}{n}\sum \max(0, 1 - y_i(w^T x_i + b))$, $\eta_t = 1/(\lambda t)$.
- Best $\lambda = 0.01$: margin 0.73, support fraction 0.40, val F1 0.832.
- RFF extension (D=500) tried, trails linear (0.767): linear kept.

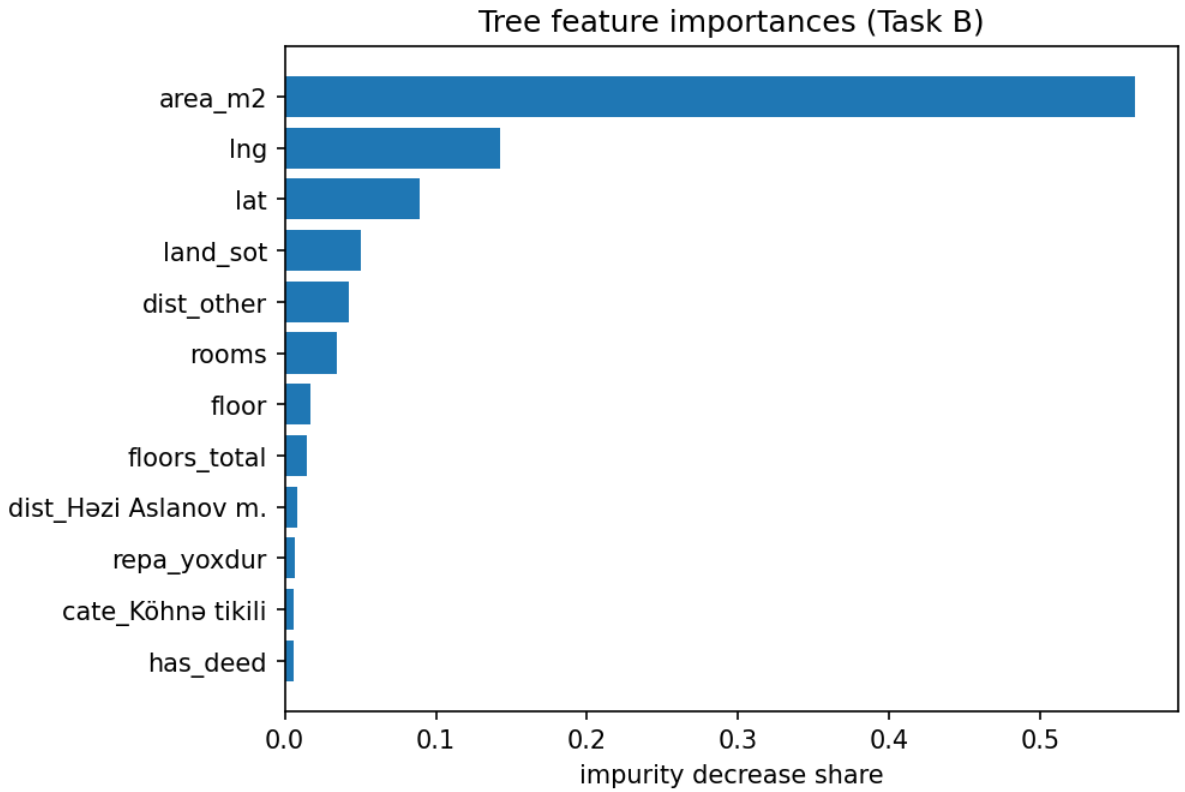


Results

Model	Task A RMSE	Task B F1
My tree	368,435	0.918
sklearn tree	377,891	0.919
My SVM	—	0.832
sklearn SGD	—	0.854

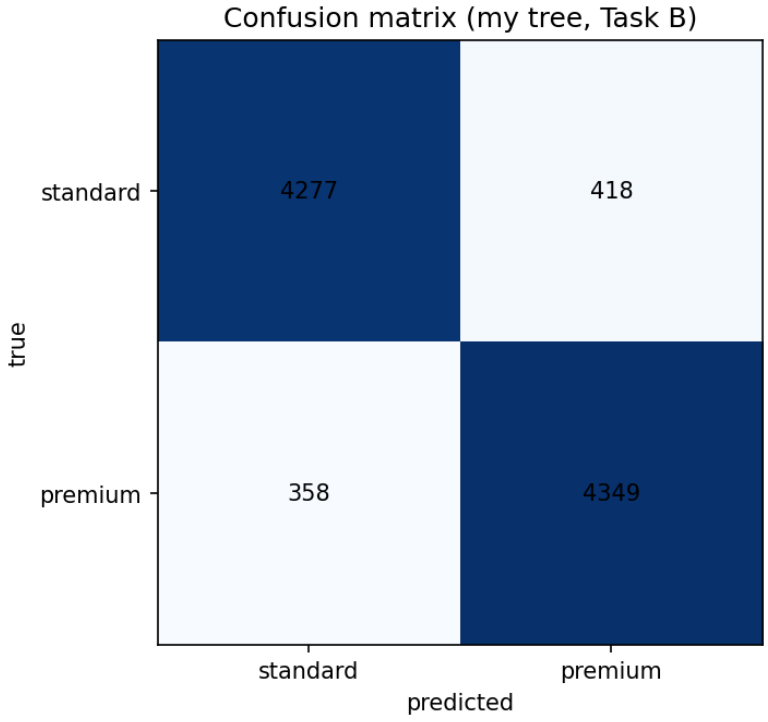
- Test: tree R^2 0.55, AUC 0.955; SVM AUC 0.920. CV: 0.911$\pm$0.002 / 0.832$\pm$0.007.
- One command reproduces all: `python -m src.run_all`.

What drives predictions



Area, district and rooms dominate; deed/repair flags help at the margin.

Where models fail



- Luxury outliers inflate RMSE (MAE only 86k); sparse outskirt districts are thin.
- Single split headline + train-only CV; no text/image features yet.

Next steps

- Description-text features and district-grouped error modelling.
- Contributions: see `contribution_report.md` in the repo root.
- Repo: `github.com/nuralnetworks/bina-az-price-prediction` (tag `v1.0-final`); report + this deck submitted on Moodle.